

Knowledge is Power



Team Crystal



Benjamin Keckley
Enjoys singing, guitar,
piano, game dev, art, cats,
and coffee.



Marshall Hoar
Enjoys biking, fishing,
and ultimate frisbee.



Paul Flynn
Likes to walk.



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Likes to argue.



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Computer science,
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Pensacola Ice Flyers #1
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Chris Humphreys
Loves cars, deranking in
valorant, and overtime.

Meet the Team

CS410 - Crystal - Knowledge Is Power

Elevator Pitch

Knowledge Is Power (KIP) is a legal literacy platform that teaches young people the laws that affect their everyday lives through online games.

Starting with Virginia traffic and copyright law, KIP aims to make legal knowledge more accessible, relevant, and easier to retain.



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Problem Statement

Young people are expected to follow laws that are often never taught in a practical way. Instead, they trust parents, friends, or the internet to inform their understanding of the law.

This can lead to confusion, misinformation, and otherwise avoidable legal consequences. KIP addresses this by teaching everyday legal concepts through guided, interactive scenarios, beginning with Virginia traffic law as the first module.



Problem Characteristics

- Young people often receive little formal education about the legal system ¹
- Legal language can be difficult to understand without prior knowledge
- Legal content must be kept accurate as laws and policies change
- Many existing resources focus on memorizing facts instead of applying them to real situations ²
- Traffic law is a practical starting point as young drivers face them early

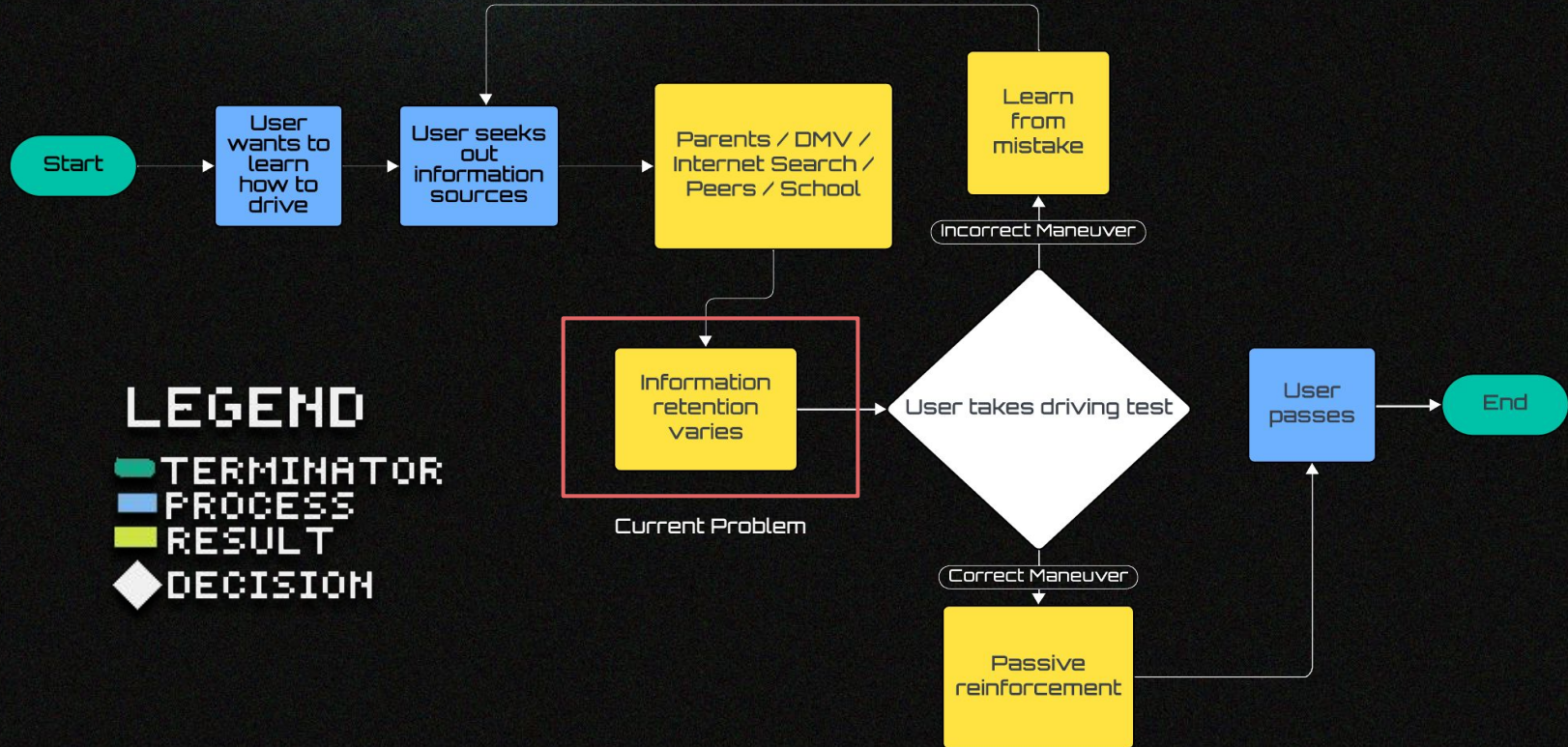


1: Touro Law Jacob D. Fuchsberg Law Center. (2017)

2: National Library of Medicine. (2022)

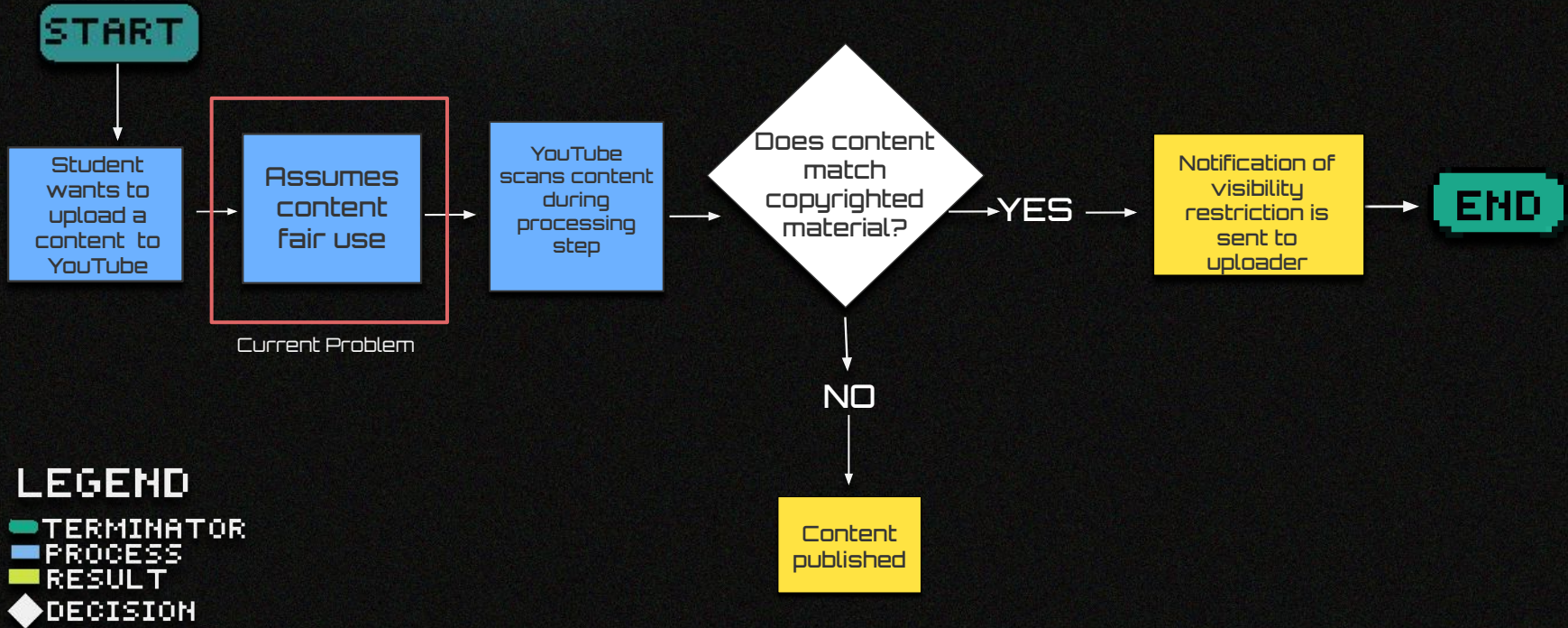
Current Process Flow

Example 1 - Driving Education and Certification



Current Process Flow

Example 2 - Copyright Infringement



Solution Statement

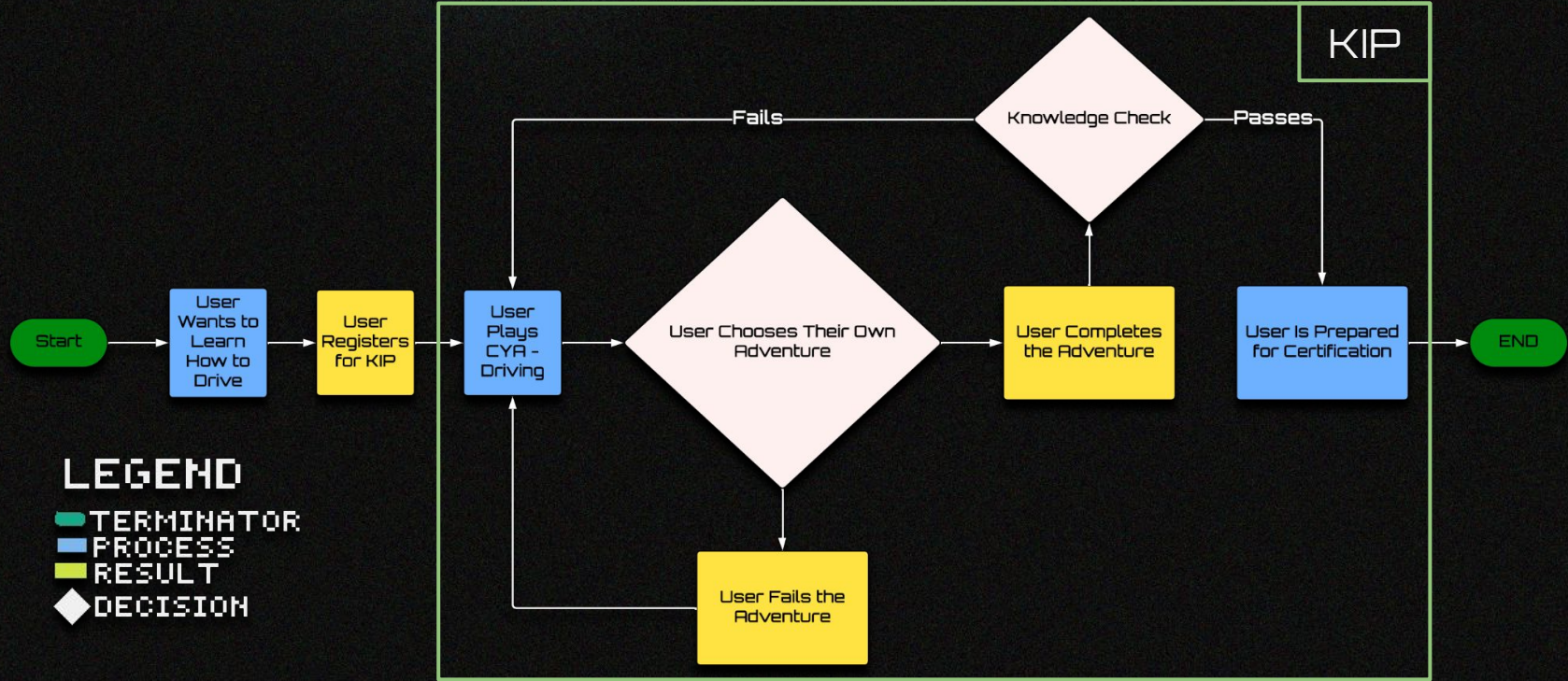
KIP will be an interactive web application that teaches practical legal literacy through game-based modules. Each module will present users with realistic legal scenarios, guide them through decision-making, and offer immediate feedback to reinforce correct understanding.

The first and second modules will focus on Virginia traffic law and copyright law, respectively, to demonstrate the platform's potential for scalability into additional areas of everyday law.



Solution Process Flow

Example 1 - Driving Education and Certification



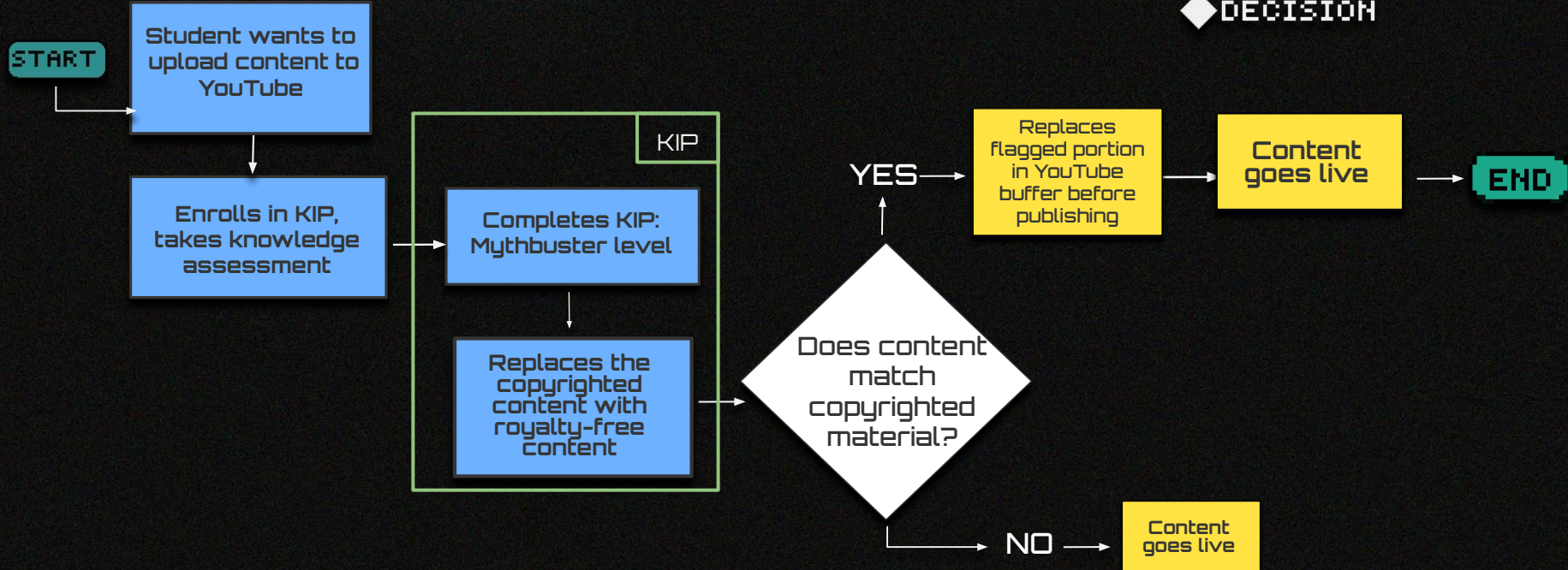
Solution Process Flow

Example 2 - Copyright Infringement



LEGEND

- TERMINATOR
- PROCESS
- RESULT
- DECISION



It will...

- ...teach and reinforce legal literacy through game-based learning
- ...help users better understand and remember commonly-applicable laws
- ...give immediate, actionable feedback on user decisions
- ...recommend and expose content gradually to prevent overwhelming new users
- ...measure comprehension to identify weaknesses and determine mastery
- ...reference original legal sources when explaining rationale for deeper learning
- ...remain scalable to support future expansion to additional modules
- ...remain publicly accessible as a web app on desktop and mobile devices



It will *not*...

- ...replace a formal education program on any module topic
- ...provide professional training in any module topic's domain
- ...require any downloads, installations, special peripherals, or advanced hardware beyond an up-to-date web browser
- ...cover laws outside of federal and Virginia state laws
- ...exhaustively cover all Virginia laws
- ...provide defensible legal advice or substitute for a licensed attorney



Competition Matrix



<i>Feature</i> \ <i>Competitor</i>	KIP	Street Law	iCivics	Law Dojo	Legal Textbooks
Generalizable Legal Literacy	Y	Y	Limited	Limited	Y
State-Specific Law Content	Y	Limited	Limited	Limited	
Gameplay-Based Learning	Y	Limited	Y	Y	
Decision-Driven Scenarios	Y	Y	Y	Limited	Limited
Immediate Explanatory Feedback	Y		Limited	Limited	
Progress-Based Recommendations	Y		Limited	Limited	



Non-Competitors

- Law firms
- Law school
- Legal databases
- Driving courses
- AI chatbots

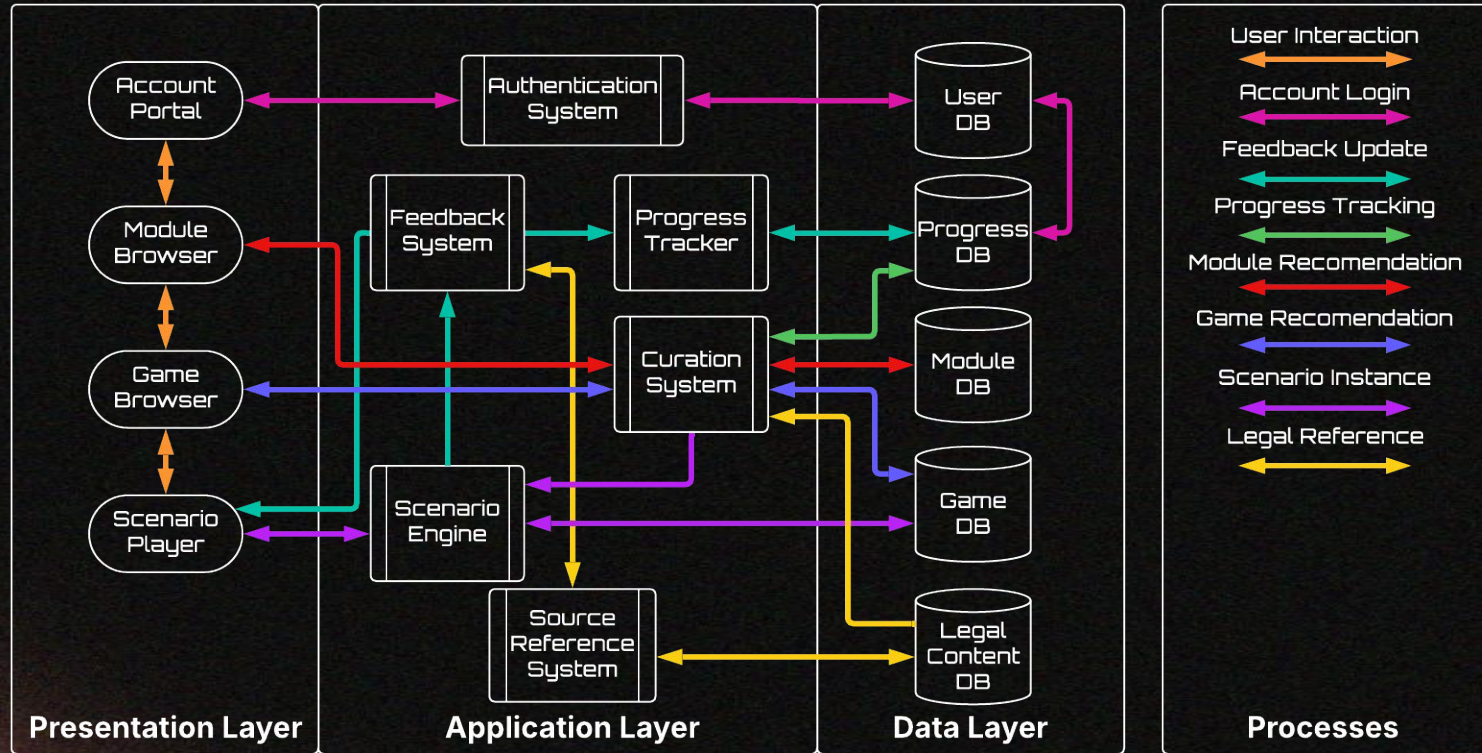


Major Functional Components

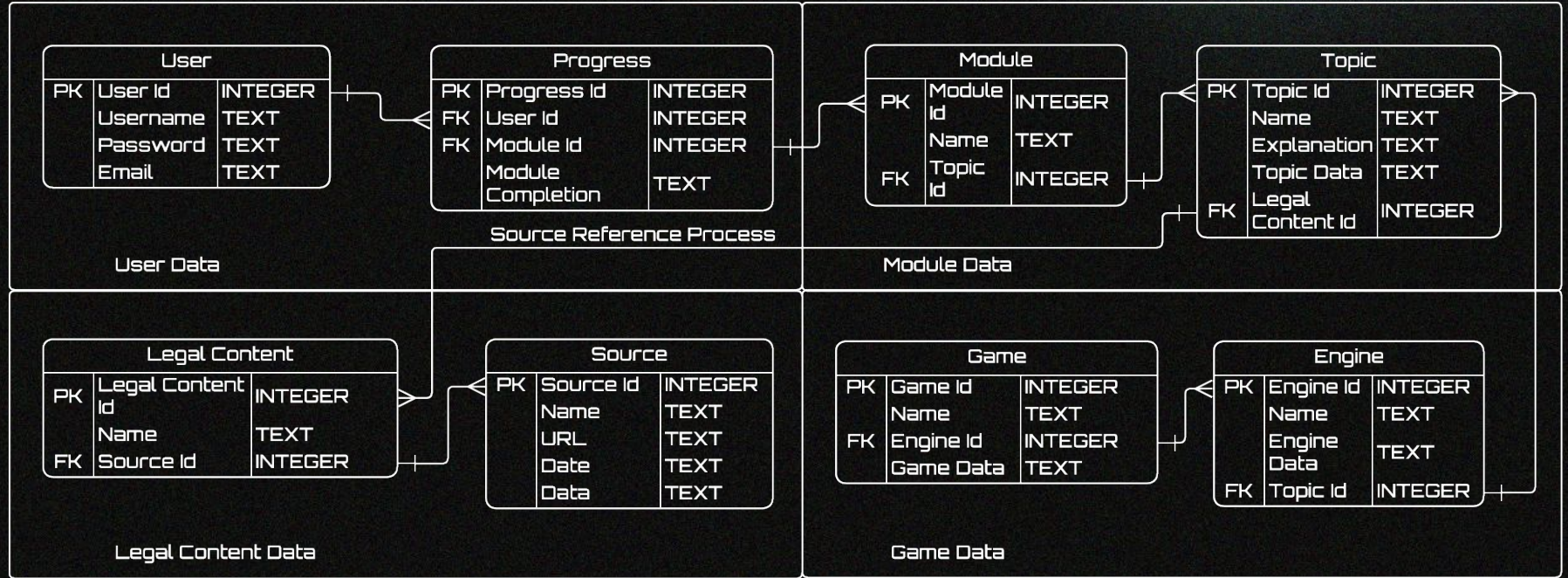
- Authentication System
- Module/Game Browser
- Scenario Engine/Player
- Feedback System
- Progress Tracker
- Curation System
- Source Reference System



Major Functional Components Diagram



Database Schema Table



Development Frameworks

What...



- Back-end: **Java**, **JDBC**, **Spring Boot**
- Front-end: **JavaScript**, **Bootstrap**, **HTML5**
- Database: **SQLite**



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How...



- Development environment: **VS Code**
- Testing framework: **JUnit**
- Documentation: **Javadoc**, **JSDoc**



Development Frameworks

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- Front-end: **JavaScript**, **Bootstrap**, **HTML5**
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How...

- Development environment: **VS Code**
- Testing framework: **JUnit**
- Documentation: **Javadoc**, **JSDoc**



Where...

- Hosting: **GitHub Pages**
- Version control: **Git**, **GitHub**



Technical Risks

Description:

Cross-device compatibility: smartphones, tablets, computers, etc. Scalability is also vital to the program's success



Probability: 4/5

Multi-platform usage is bound to have issues



Impact: 5/5

Technical issues limit information retention and pushes users away.

Mitigation:

Cloud based hosting can largely solve any scalability issue, and browser testing can mitigate much of the cross device compatibility problems.



Probability: 2/5

A unilateral access point aids in data and user access



Impact: 3/5

In the case of a security breach backups will aid in system recovery

Legal Risks - Copyrighted Content



Description:

KIP may use real-world legal scenarios and publicly available information. Improper use of copyrighted material could lead to legal issues.



Probability: 3/5

Legal language from official law documents difficult to understand



Impact: 5/5

Inaccuracy causes mistrust, legal conflicts stops production

Mitigation:

Use only public domain and open-source legal databases. Adhering to CC By Attribution Policy, implementing the rigorous citation protocol: TASL (Title, Author, Source License).



Probability: 2/5

Slight mitigation through diligent monitoring of changes to laws



Impact: 5/5

Promoting accountability and diligence

Legal Risks - Obsolete Information



Description:

As laws evolve over time, some modules have the potential to teach obsolete material, rendering KIP untrustworthy.



Probability: 4/5

Requires strict updates to keep legal information up to date



Impact: 5/5

Outdated information affects user trust

Mitigation:

Date-modified checks, prominent legal disclaimer, and modular design for efficient content updates.



Probability: 1/5

Reduces passive liability through proactive design and clear responsibility boundaries



Impact: 5/5

Impact still crucial, but risk lowered

Algorithm 1: Legal Source Maintenance

KIP maintains references to legal sources to avoid offering scenarios with answers, explanations, or feedback that may be obsolete or unreliable.

At scheduled intervals, a maintenance algorithm evaluates whether any referenced legal sources have changed.

1. Fetch all legal sources referenced by active scenarios.
2. For each legal source, check whether it has been updated, repealed, replaced, or otherwise changed since the last scan.
3. Identify all scenarios that depend on one or more changed legal sources.
4. Temporarily flag affected scenarios as requiring review and remove them from the active scenario pool.
5. Notify administrators of the flagged scenarios.

Security Risks

Description:

Data privacy and unauthorized access are always a security risk when it comes to software applications



Probability: 3/5

Without the right tools it takes one bad actor to do serious damage



Impact: 5/5

Sensitive information would be available for disruption.

Mitigation:

KIP will mitigate this by encrypting all stored data and implementing role based security measures. System backups ensure an easily recoverable state.



Probability: 2/5

Encryption and role-based security will limit the methods of attack



Impact: 3/5

In the case of a security breach backups will aid in system recovery

User/Adoption Risks



Description:

Young people may lose interest if the educational content is not properly balanced, and gamification may overshadow primary law content.



Probability: 4/5

Legal language is complex and foreign to many young people



Impact: 5/5

Success relies on engagement from beginning to end

Mitigation:

KIP will accurately assess current knowledge level. Knowledge checks give users a goal to work towards. Focus groups and acceptance tests helps to find the right balance.



Probability: 1/5

Gamification and focus groups will simplify dense terms



Impact: 5/5

Testing phase catches engagement vulnerabilities

Algorithm 2: Curated Scenario Selection

KIP prepares games using a priority-based selection algorithm.

This algorithm considers 3 major factors:

- Which topics the game should test.
- The user's proficiency in each assessed topic.
- How recently the user has encountered each scenario.

1. Retrieve the full set of topics assessed by the selected game.
2. For each topic, retrieve the user's current proficiency score.
3. Identify their weakest topics by selecting the lowest-scored topics.
4. Retrieve all scenarios that assess 1 or more weakest topics.
5. Order the matching scenarios from least recently played to most recently played.
6. Choose the set of oldest scenarios that cover the least proficient topics.

Work Breakdown Structure

User Interface	Backend & Application Logic	Database Management	Algorithms	Unit & Integration Testing
Homepage design Module selection interface Scenario gameplay scenes Feedback & results display	Server logic Scenario engine Game flow control Authentication Progress tracking logic Spring Boot integration	SQLite database management Module data User data Legal content data Game data JDBC integration	Legal source maintenance Curated scenario selection	Unit testing (JUnit) Integration testing System testing Debugging

Real-World Product VS Prototype Table



Features & Functionality	RWP	Prototype (Planned)	Prototype (Actual)
United States Traffic Law	Coverage of Federal and all fifty states	Virginia traffic only	TBD
Copyright Law	Comprehensive copyright law	Social media platforms	TBD
Adaptive Game Difficulty	Relaxed, Standard, Intense modes	Level 1 (Relaxed) Level 2 (Standard) Level 3 (Intense)	TBD
User Account and History	Server-side database	SQLite	TBD

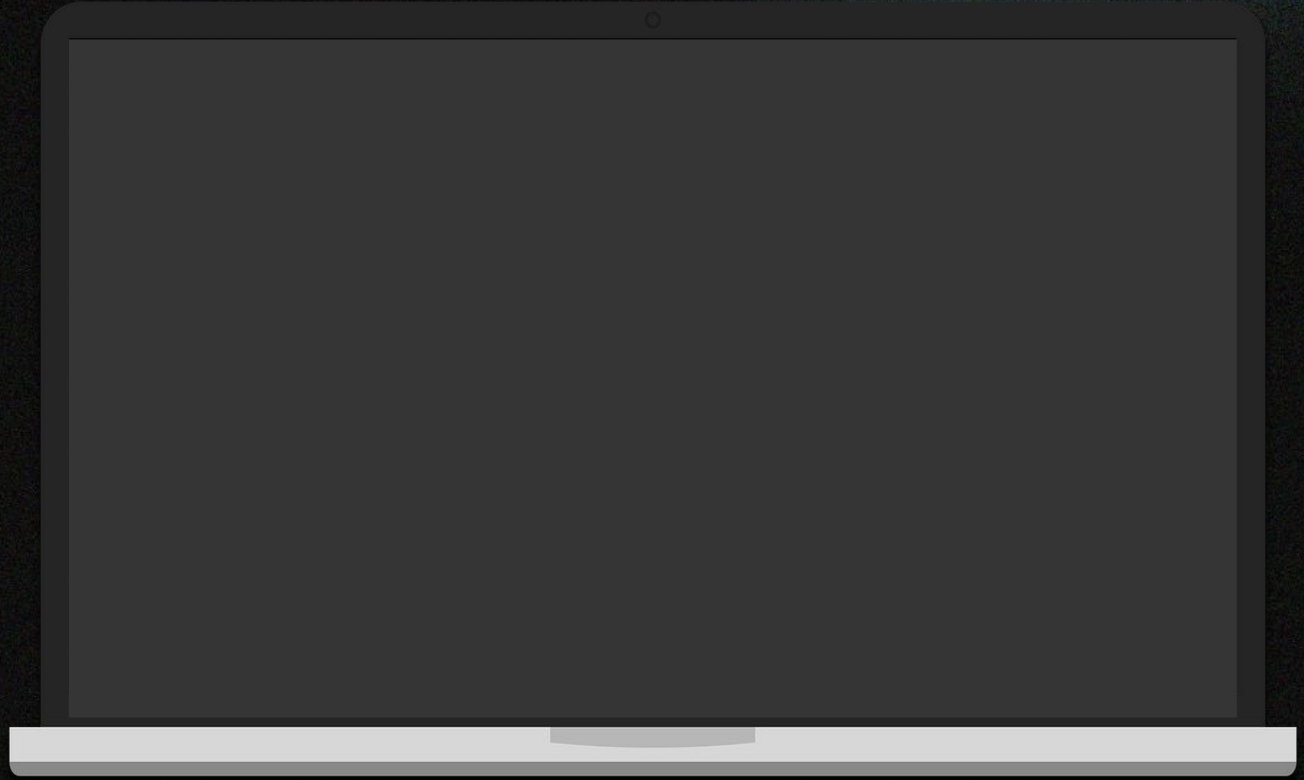


Prototype Roadmap 🔊

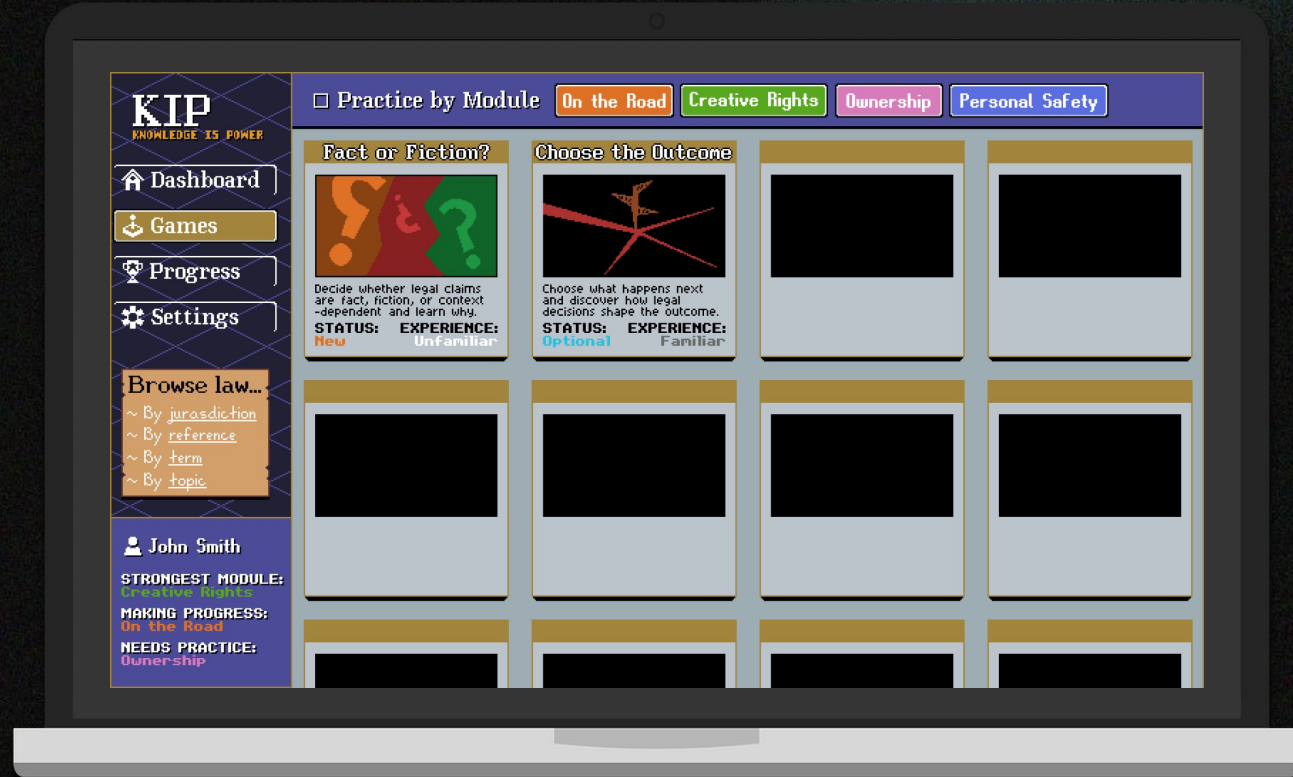
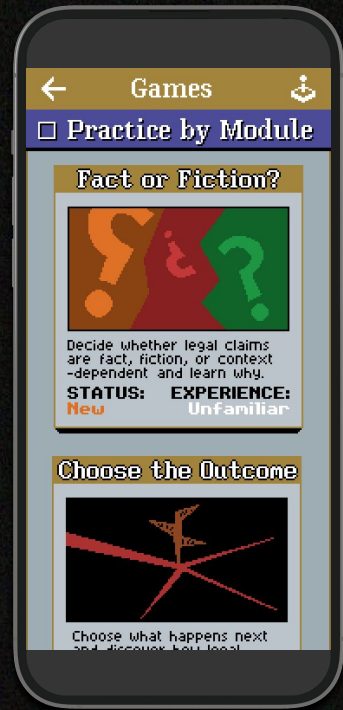
Features & Functionality	Prototype (Planned)
United States Traffic Law	Hardcode mock data sandboxed to Virginia
Copyright Law	Regex filtering to check legal accuracy
Adaptive Game Difficulty	Level 1 (Relaxed) hardcoded to transition to level 2 (Standard)
User Account and History	Tracking retention and engagement metrics via SQLite and JDBC



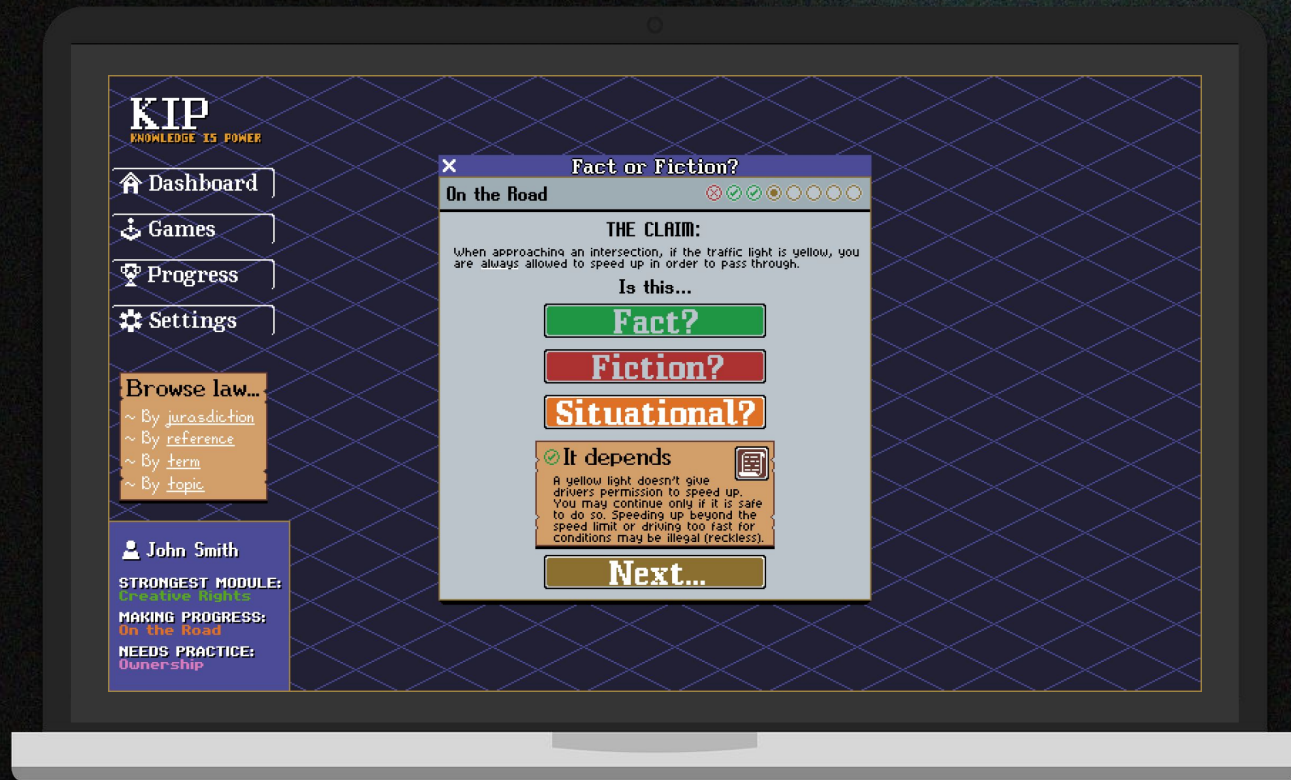
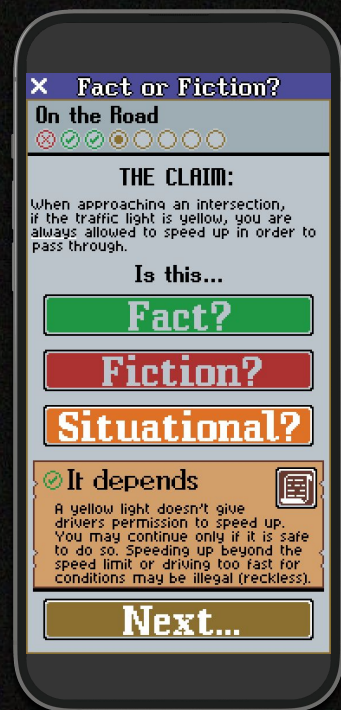
GUI Mockup 1: Your Dashboard



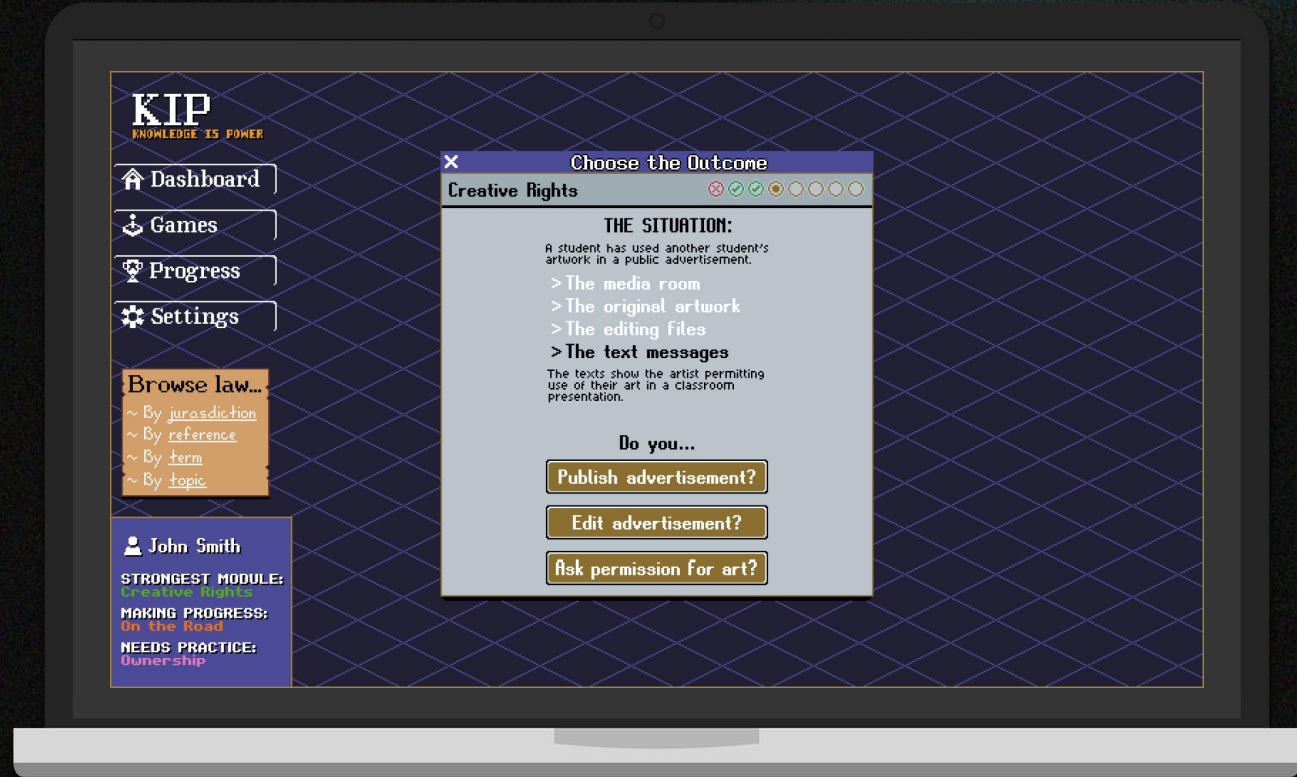
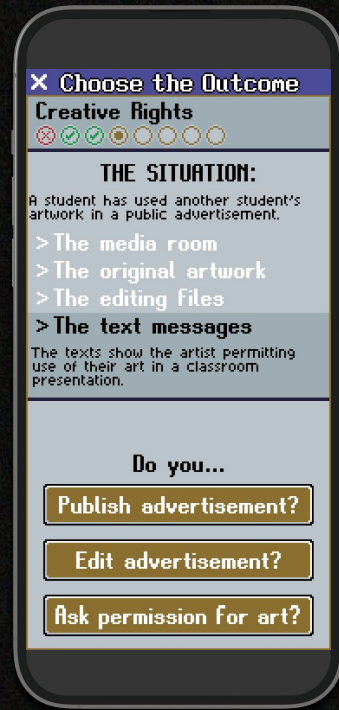
GUI Mockup 2: Game Browser



GUI Mockup 3: “Fact or Fiction?” Game



GUI Mockup 4: “Choose the Outcome” Game



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Appendix



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- Slide [33] u_83ynl5kta9. (2026). *Objection sound effect* [Audio clip]. Pixabay. <https://pixabay.com/sound-effects/>